

Total No. of Questions : 10]

SEAT No. :

P3228

[Total No. of Pages : 2

[5354]-684

B.E. (Computer Engineering)

MOBILE COMPUTING (Elective - III)

(2012 Pattern)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8 and Q9 or Q10
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Assume suitable data if necessary.

Q1) a) What is the need of spread spectrum technology? Explain DSSS with one example [5]

b) Describe mobile computing architecture with neat diagram. [5]

OR

Q2) a) Differentiate between different wireless switching techniques. [5]

b) Explain in detail GSM network architecture. [5]

Q3) a) Explain with a neat diagram GSM network interfaces and protocols. [5]

b) Compare TDMA and FDMA systems. [5]

OR

Q4) a) What is handover? When handover in mobile communication takes place? [5]

b) How the next generation wireless technologies are different than 3G technologies. [5]

Q5) a) What is basic purpose of DHCP? Name the entities of DHCP. [6]

b) Explain tunneling and encapsulation in mobile IP. [6]

c) Explain in detail the differences between AODV and the standard distance vector algorithms. [6]

OR

P.T.O.

- Q6)** a) What is tunneling ? How does a tunnel differ from a route? [6]
b) Explain Snooping TCP. Give the advantages and disadvantages of Snooping TCP. [6]
c) What are the security threats to a MANET ? Why a MANET faces greater security threats than a fixed infrastructure network? [6]

- Q7)** a) Describe pull-based data-delivery mechanism in detail . What are the advantages and disadvantages of pull-based data-delivery? [8]
b) What are the different types of data synchronizations in mobile computing systems? Describe synchronization usage models in mobile applications. [8]

OR

- Q8)** a) What is selective tuning? Discuss the methods used for selective tuning. [8]
b) Explain what is synchronization protocol? Explain how Bluetooth synchronizes mobile applications. [8]

- Q9)** a) What is mobile agent? What are advantages of using a mobile agent. [8]
b) Write Short note on [8]
i) Difference between iOS and Android OS.
ii) Service discovery and device management in mobile O.S.

OR

- Q10)** a) Explain what is gateway? Describe a transcoding gateway and its applications in mobile computing systems. [8]
b) Write Short note on [8]
i) Functions of Application Server.
ii) Requirement of Mobile File System.

